Assignment

Q1. Write a C program for calculating the price of a product after adding the sales tax to its original price. Where rate of tax and price is inputted by user.

A1.

```
 #include <stdio.h>
#include <conio.h>

int main()
{
    float op,p,r,st;

    printf("enter the original price and rate respectively: ");
    scanf("%f%f",&op0,&r);

    st= op*(r/100);
    p= st+op;

    printf("the final price of the product after adding the sales tax is: %.2f",p);
    return 0;
}
```

Q2. Write a C program to calculate the weekly wages of an employee. The pay depends on wages per hour and number of hours worked. Moreover, if the employee has worked for more than 30 hours, then he or she gets twice the wages per hour, for every extra hour that he or she has worked.

A2.

```
#include <stdio.h>
#include <conio.h>

int main()
{
    float w,hw,wph;

    printf("enter the hours worked by the employee and the wage provided per hour respectively: ");
    scanf("%f%f",&hw,&wph);

    if(hw<=30 && hw>=0)
    {
        w=hw*wph;
        printf("the wage rewarded to the employee for working %.2f hours is %.2f",hw,w);
    }
    else if(hw>30)
    {
        w=hw*(wph*2);
        printf("the wage rewarded to the employee for working %.2f hours is %.2f",hw,w);
    }
    else
    {
        printf("invalid input");
    }


    return 0;
}
```

Q.3 Mr. X goes to market for buying some fruits and vegetables. He is having a currency of Rs 500 with him for marketing. From a shop, he purchases 2.0 kg Apple priced Rs. 50.0 per kg, 1.5 kg Mango priced Rs.35.0 per kg, 2.5 kg Potato priced Rs.10.0 per kg, and 1.0 kg Tomato priced Rs.15 per kg. He gives the currency of Rs. 500 to the shopkeeper. Find out the amount shopkeeper will return to X by writing a C program.

A3.

```
#include <stdio.h>
#include <conio.h>
```

```
int main()
{
    float tc=500,a=2,ra=50,m=1.5,ma=35,p=2.5,pa=10,t=1,ta=15,ad,cl;

    ad= (a*ra)+(m*ma)+(p*pa)+(t*ta);
    cl = tc-ad;

    printf("the shopkeeper will return %.2f rs. to Mr.X ",cl);

    return 0;
}
```

Q4.Write a C program to print your name, date of birth and mobile number in 3 different lines.

A4.

```
#include <stdio.h>
#include <conio.h>

int main()
{
    printf("My name is January.\nMy date of birth is 01/01/2001.\nMy mobile number is 1234567890.");

}
```

Q5.Write a program to read an integer, a character and a float value from keyboard and display the same in different lines on the screen.

A5.

```
#include <stdio.h>
#include <conio.h>

int main()
{
    float f;
    int i;
    char c;
    printf("Enter the float, integer and character value respectively:");
    scanf("%f%d%c",&f,&i,&c);

    printf("the float value is %f, the integer value is %d and the character value is %c.");
}
```

Q6.Write a program to print the following line (Assume the total value is contained in a variable named cost)
The sales total is : $ 172.53

A6.

```
#include<stdio.h>
#include<conio.h>

int main()
{
    float cost = 172.53;

    printf("the sales total is: $%.2f",cost);
    return 0;
}
```

Q7.Raju got 6 and half apples from each of Raghu, Sheenu and Akash. He wants to know how many apples he has in total without adding them. Write a program which could help Raju in doing this.

A7.

```
#include <stdio.h>
#include<conio.h>
```

```
int main()
{
 int Raghu = 6.5;
 float Sheenu = 6.5;
 int Akash = 6.5;
 float ta =Raghu +Sheenu + Akash;
 printf("Raju has a total of %.1f apples.\n", ta);
 return 0;
}
```

Q8.Write a program that prints the floating point value in exponential format correct to two decimal places.

A8.

```
#include <stdio.h>
#include<conio.h>
int main()
{
    float f;

    scanf("%f",&f);

    printf("in float: %.2f \n",f);
    printf("in expopnent: %.2e",f);

    return 0;
}
```

Q9.Write a program to input and print your mobile number (i.e. of 10 digits).

A9.

```
#include<stdio.h>
#include<conio.h>

int main()
{
    long int num;

    printf("enter the phone number: ");
    scanf("%ld",&num);

    printf("the phone number is: %ld",num);
    return 0;
}
```

Q10.The population of a city is 30000. It increases by 20 % during first year and 30% during the second year. Write a program to find the population after two years? (Ans: 46800)

A10.

```
#include <stdio.h>
int main()
{
 int initialP = 30000;
 float Year1 = 0.20;
 float Year2 = 0.30;

 float population1 = initialP + (initialP * Year1);
 float population2 = population1 + (Year1 * Year2);

 printf("The population after two years is: %.0f\n", Year2);
```

```
 return 0;
}
```

Q11. Write a program to find the ASCII value of a character.

A11.

```
#include <stdio.h>
#include <conio.h>

int main()
{
 char c;

 printf("Enter a character: ");
 scanf("%c", &c);

 printf("The ASCII value of '%c' is %d\n", c, c);

 return 0;
}
```

Q12. Write a program to calculate salary of an employee, given his basic pay (entered by user), HRA=15% of the basic pay and TA=20% of the basic pay.

A12.

```
#include <stdio.h>
#include <conio.h>
int main()
{
 float basicPay, HRA, TA, salary;

 printf("Enter the basic pay of the employee: $");
 scanf("%f", &basicPay);

 HRA = 0.15 * basicPay;
 TA = 0.20 * basicPay;
 salary = basicPay + HRA + TA;

 printf("Salary of the employee is: $%.2f\n", salary);

 return 0;
}
```

Q13. Write a program to find the slope of a line and angle of inclination that passes through two points P and Q with coordinates (xp, yp) and (xq, yq) respectively.

A13.

```
#include <stdio.h>
#include <math.h>

int main()
{
 float xp, yp, xq, yq, slope, angle;
```

```
 printf("Enter the coordinates of point P (xp yp): ");
 scanf("%f %f", &xp, &yp);

 printf("Enter the coordinates of point Q (xq yq): ");
 scanf("%f %f", &xq, &yq);

 slope = (yq - yp) / (xq - xp);
 angle = atan(slope) * 180 / M_PI;

 printf("Slope of the line: %.2f\n", slope);
 printf("Angle of inclination: %.2f degrees\n", angle);

 return 0;
}
```

Q14. The SPI (Semester Performance Index) is a weighted average of the grade points earned by a student in all the courses he registered for in a semester. If the grade points associated with the letter grades awarded to a student are g1, g2, g3,…….gk etc. and the corresponding credits are c1, c2, c3,…...ck, the SPI is given by:

$$\mathrm{SPI} = \frac{\sum_{i=1}^{k} c_i g_i}{\sum_{i=1}^{k} c_i}$$

Where, k is the number of courses for which the candidate remains registered for during the semester/ trimester. Write a program in C to calculate SPI for k =5.

A14.

…..

Q 15. Write a program to calculate the frequency (f) of a given wave with wavelength (λ) and speed (c), where c=λ*f.

A15.

```
#include <stdio.h>
#include <conio.h>

int main()
{
 float wavelength, speed, frequency;

 printf("Enter the wavelength (in meters): ");
 scanf("%f", &wavelength);

 printf("Enter the speed of the wave (in meters/second): ");
 scanf("%f", &speed);

 frequency = speed / wavelength;
 printf("Frequency of the wave: %.2f Hz\n", frequency);

 return 0;
}
```

Q 16. A car travelling at 30 m/s accelerates steadily at 5 m/s2 for a distance of 70 m. What is the final velocity of the car? [Hint: v2 = u2 + 2as]

A16.

```
#include <stdio.h>
#include <math.h>
```

```
int main()
{
 float initialVelocity = 30.0,acceleration = 5.0,distance = 70.0,finalVelocity;

 finalVelocity = sqrt(initialVelocity * initialVelocity + 2 * acceleration *
distance);
 printf("Final velocity of the car: %.2f m/s\n", finalVelocity);

 return 0;
}
```

Q 17.A horse accelerates steadily from rest at 4 m/s2 for 3s. (a) What is its final velocity? (b) How far has it travelled? [Hint: (a) v = u + at (b) s = ut + ½at2]

A17.

```
#include <stdio.h>
int main()
{
 int rollNumber = 12345678; // Replace with your university roll number
 int lastFourDigits, sum = 0;
 lastFourDigits = rollNumber % 10000;
 while (lastFourDigits > 0) {
 sum += lastFourDigits % 10;
 lastFourDigits /= 10;
 }
 printf("Sum of the last four digits of the roll number: %d\n", sum);
 return 0;
}
```

Q 18. Write a program to find the sum of your four last digit of your university roll number .

A18.

```
#include <stdio.h>
int main() {
 int rollNumber = 12345678; // Replace with your university roll number
 int lastFourDigits, sum = 0;
 lastFourDigits = rollNumber % 10000;
 while (lastFourDigits > 0) {
 sum += lastFourDigits % 10;
 lastFourDigits /= 10;
 }
 printf("Sum of the last four digits of the roll number: %d\n", sum);
 return 0;
}
```

Q19. Write a program to initialize your height and weight in cm. and kgs respectively demonstrating compile time initialization and convert them in feets and pounds respectively. **Note :- 1 cm = 0.393701inch , 1 Kg = 2.20462**

A19.

```
#include <stdio.h>
int main() {
 float heightInCm = 175.0; // Replace with your height in centimeters
 float weightInKgs = 70.0; // Replace with your weight in kilograms
 float cmToInch = 0.393701;
```

```
 float kgToPound = 2.20462;
 float heightInFeet = heightInCm * cmToInch / 12.0;
 float weightInPounds = weightInKgs * kgToPound;
 printf("Height in feet: %.2f feet\n", heightInFeet);
 printf("Weight in pounds: %.2f pounds\n", weightInPounds);
 return 0;
}
```

Q 20 . Code the variable declarations for each of following:

a) A character variable named option.
b) An integer variable sum initialized to 0
c) A floating point variable, product, initialized to 1

A20.

```
char option;
int sum = 0;
float product = 1.0;
```

Q21. Write a program that reads nine integers. Display these numbers by printing three numbers in a line separated by commas.

A21.

Q22. What are header files and what are its uses in C programming?

A22.

```
Header files in C contain declarations of functions, variables, and macros that are
used in your
program. They allow you to include pre-written code and libraries in your program.
They are
included using #include directives.
```

Q23. What will be the output of following program?

```
#include<stdio.h>
int main()
{   int num=070;
printf("%d\t%o\t%x",num,num,num);
}
```

A23.

```
The output of the given program is "56 70 38".
```

Q 24. What will be the output of following program?

```
#include <stdio.h>
void main()
{
int x = printf("GLA UNIVERSITY");
printf("%d", x);
 }
```

A24.

```
The output of the program is "GLA UNIVERSITY12.
```

Q25. What are library functions? List any four library functions.

A25.

```
Library functions in C are pre-written functions provided by the C standard library
that can be used
```

```
to perform common tasks. Four examples of library functions are printf(), scanf(),
strlen(), and sqrt()
```

Q26. What will be the output of following program?

```
#include <stdio.h>
void main()
{
  int x = printf("C is placement oriented Language") – printf("Hi");
  printf("%d %o %x", x,x,x);
}
```

A26.

```
The output of the program is "28 34 1c".
```

Q27. What is the meaning of following statement? printf("%d",scanf("%d%d",&a,&b));

A27.

```
 The scanf function returns the number of input items successfully matched and
assigned. So,
printf("%d", scanf("%d%d", &a, &b)); will print the number of successfully scanned
integers.
```

Q28. What will be the output of following program?

```
#include <stdio.h>
void main()
{
  printf(" \"C %% FOR %% PLACEMENT\"");
}
```

A28.

```
The output of the program is " "C % FOR % PLACEMENT""
```

Q29. Suppose distance between GLA University and Delhi is m km (to be entered by user), by BUS you can reach Delhi in 4 hours. Develop a 'C' program to calculate speed of bus.

A29.

```
#include <stdio.h>
int main() {
 double distance;
 double time = 4.0; // 4 hours
 printf("Enter the distance in kilometers: ");
 scanf("%lf", &distance);
 double speed = distance / time;

 printf("The speed of the bus is %.2lf km/h\n", speed);
 return 0;
}
```

Q30. In an exam Satyam got 50 marks, Suman got 70 marks and Shyam got 80 marks, Write a 'C' program to find average marks of these three participants.

A30.

```
#include <stdio.h>
int main() {
 int satyam_marks = 50;
 int suman_marks = 70;
 int shyam_marks = 80;
 int total_marks = satyam_marks + suman_marks + shyam_marks;
```

```
 double average_marks = (double)total_marks / 3.0;
 printf("Average marks of the three participants: %.2lf\n", average_marks);
 return 0;
}
```

Q31. One day, Mohan called Saurav and Sajal and gave some money to them, later he realized that money that was given to Saurav should be given to Sajal and vice-versa. Develop a 'C' program to help Mohan so that he can rectify his mistake.

A31.

```
#include <stdio.h>
int main() {
 int mohan_money = 100; // Initial amount given to Mohan
 int saurav_money = 50; // Initial amount given to Saurav
 int sajal_money = 75; // Initial amount given to Sajal
 // Swapping the money between Saurav and Sajal
 int temp = saurav_money;
 saurav_money = sajal_money;
 sajal_money = temp;
 printf("Mohan rectified the mistake.\n");
 printf("Mohan now has %d rupees.\n", mohan_money);
 printf("Saurav now has %d rupees.\n", saurav_money);
 printf("Sajal now has %d rupees.\n", sajal_money);
 return 0;
}
```

Q32. One day when I was going for a lunch, suddenly rain started, I was very hungry so started running with speed of 4km/h and it took 3 min to reach mess. Help me to develop a 'C' program to calculate distance travelled by me.

A32.

```
#include <stdio.h>
int main() {
 float speed_kmph = 4.0; // Speed in km/h
 float time_min = 3.0; // Time in minutes
 // Convert time to hours
 float time_hours = time_min / 60;
 // Calculate distance
 float distance_km = speed_kmph * time_hours;
 printf("Distance traveled by you: %.2f kilometers\n", distance_km);
 return 0;
}
```

Q33. Can two or more escape sequences such as \n and \t be combined in a single line of program code?

A33.

```
 Yes, you can combine multiple escape sequences in a single line of program code
```

Q34. What are comments and how do you insert it in a C program?

A34.

```
Comments in C are used to provide explanations and descriptions within the source
code. They
are not executed by the compiler. Comments in C can be inserted using /* */ for
multiline
comments and // for single-line comments
```

Q35. What is wrong in this statement? scanf("%d",number);

A35.

```
 The statement scanf("%d", number); is incorrect because it should be scanf("%d",
&number); to correctly read an integer and store it in the variable number.
```

Q36. What will be the output?

```
#include <stdio.h>
     int main()
{
   if (sizeof(int) > -1)
      printf("Yes");
   else
      printf("No");
   return 0;
}
```

A36.

```
The output of the program is "Yes."
```

Q37. Point out which of the following variable names are invalid:
gross-salary INTEREST , salary of emp , avg. , thereisbookinmysoup
A37.

```
 Variable names "gross-salary," "INTEREST," "avg," and "thereisbookinmysoup" are
valid in C.
```

Q38. Tom works at an aquarium shop on Saturdays. One Saturday, when Tom gets to work, he is asked to clean a 175-gallon reef tank. His first job is to drain the tank. He puts a hose into the tank and starts a siphon. Tom wonders if the tank will finish draining before he leaves work. He measures the amount of water that is draining out and finds that 12.5 gallons drain out in 30 minutes. So, he figures that the rate is 25 gallons per hour. Develop a 'C' program to help Tom to calculate time required to completely clean tank.

A38.

```
To calculate the time required to completely clean the tank, you can use the
formula: time =
(tank_capacity / drain_rate), where tank_capacity is 175 gallons and drain_rate is
25 gallons per hour
```

Q39. The percent y (in decimal form) of battery power remaining x hours after you turn on a laptop computer is y = −0.2 x + 1. Develop a 'C' program to calculate after how many hours the battery power is at 75%?
A39.

```
 To calculate the time when the battery power is at 75%, you can use the formula: x
= (1 - y)
/ (-0.2), where y is 0.75 (75%).
```

Q40.Which of the following is used to convert the high level language in machine language in a single go?
a. Compiler b.Interpreter
c. Linker d.Assembler
A40. Assembler.

Q 41. What is the format specifier for an Octal Number?
a.%0 b.%d
c. %o d. %e
A41. %o.

Q 42. Which format specifier is used to print the exponent value upto 2 decimal places.
a. %e b.%.2f c. %f d.%.2e
A42. %e.

Q 43. Which of the following is not a basic data type?
a. char
b. array
c. float
d. int
A43. Array.

Q 44. What is the output of following code?

```
#include<stdio.h>
void main()
{
```

```
  int x=0;
  x= printf("\"hello\b\"");
  printf("%d",x);
}
```

a. hello7 b. "hello"7 c. "hell"8 d. hell8

A44. "hell"8

Q 45. What is the output of following code?

```
#include<stdio.h>
void main()
{
  int b,c=5 ;
  int("%d , %d", b,c);
}
```

a. 5, 5 b. 5, 5.000000

c. Garbage, 5.000000 d. Garbage, 5

A45. Garbage, 5

Q46. Which of the following is an identifier?

a. &fact b. Basic_pay c. enum d. 1sum

A46. Basic_pay

Q 47. What is the output of the following program?

```
#include<stdio.h>
void main()
{
   char x, a='c';
   x=printf("%c",a);
   printf("%d",x);
}
```

a. c1 b. cgarbage

c. 1 c. c

A47. c1

Q48. Perform the following conversion from Decimal to other number as directed-

a) $(365.55)_{10} = (?)_2$
b) $(453.65)_{10} = (?)_8$
c) $(5164.12)_{10} = (?)_{16}$
d) $(23.65)_{10} = (?)_5$
e) $(772)_{10} = (?)_7$

A48.

Q49. Covert the following numbers to decimal number system-

a) $(325.54)_6 = (?)_{10}$
b) $(1001010110101.1110101)_2 = (?)_{10}$
c) $(742.72)_8 = (?)_{10}$
d) $(AC94.C5)_{16} = (?)_{10}$

A49.

Q50. Perform the following conversion from Hexadecimal to other number as directed-

$(DB56.CD4)_{16} = (?)_2, \quad (?)_8, \quad (?)_4$

A50.

Q51. Perform the following conversion from octal to other number as directed-

$(473.42)_8 = (?)_2, \quad (?)_{10}, \quad (?)_{16}, \quad (?)_5$

A51.

Q52. Find the value of A?

a) $(23)_{10} = (17)_A$
b) $(21)_{16} = (41_{)A}$
c) $(32)_8 = (101)_A$

A52.

Q53: What will be the output of following program? Assume integer is of 2 bytes

```
void main(){
int  a=32770;
```

```
printf(“%d”,a);
}
```

A53. -32766

Q54:
```
#include <stdio.h>
 int main()
{
 float c = 5.0;
 printf ("Temperature in Fahrenheit is %.2f", (9/5)*c + 32);
 return 0;
}
```

A54.Temperature in farenheit is 41.00.